Bridgeland Stability Conditions

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1. Stability conditions on an abelian category

1.1. **Conventions.** Throughout, $\mathcal{A}$ is an essentially small abelian category and $K_0(\mathcal{A})$ is its Grothendieck group, i.e. the abelian group generated by the classes $[E]$ of objects $E \in \mathcal{A}$ subject to the relations $[E] = [E'] + [E'']$ for every short exact sequence $0 \to E' \to E \to E'' \to 0$. For an object E and a group homomorphism $Z\colon K_0(\mathcal{A}) \to \mathbb{C}$ we write $Z(E) := Z([E])$.

Let

$$\mathbb{H} := \left\{ re^{i\pi\varphi} \middle| \, r > 0,\ \varphi \in (0,1] \right\} = \{ z \in \mathbb{C} | \operatorname{Im} z > 0 \} \cup \mathbb{R}_{<0}$$

be the upper half-plane together with the negative real ray. For $z \in \mathbb{H}$ the number $\varphi \in (0,1]$ with $z \in \mathbb{R}_{>0} e^{i\pi\varphi}$ is uniquely determined; we call it the *phase* of z. Note that $\mathbb{H}$ is closed under addition and under multiplication by positive real numbers.

The notation $F \subset E$ and $E \twoheadrightarrow G$ always refers to a subobject and a quotient object in the abelian category $\mathcal{A}$. A subobject $F \subset E$ is *proper* if $F \neq E$ and *nontrivial* if $F \neq 0$.

1.2. **Stability functions.** A group homomorphism $Z\colon K_0(\mathcal{A}) \to \mathbb{C}$ is called a *stability function* on $\mathcal{A}$ if

(Z) $Z(E) \in \mathbb{H}$ for every nonzero object $E \in \mathcal{A}$.

Equivalently: $\operatorname{Im} Z(E) \geqslant 0$, and $\operatorname{Im} Z(E) = 0$ implies $\operatorname{Re} Z(E) < 0$.

Let Z be a stability function. For a nonzero object $E \in \mathcal{A}$ the *phase* of E is the number $\varphi(E) = \varphi_Z(E) \in (0,1]$ equal to the phase of the complex number $Z(E) \in \mathbb{H}$, so that

$$Z(E) = m(E)\, e^{i\pi\varphi(E)}, \qquad m(E) := |Z(E)| > 0.$$

The number $m(E)$ is called the *mass* of E.

Remark. In terms of the generalized rank $R := \operatorname{Im} Z$ and the generalized degree $D := -\operatorname{Re} Z$, the phase corresponds to the generalized slope $M := D/R = -\cot(\pi\varphi) \in \mathbb{R} \cup \{+\infty\}$, with $M = +\infty$ for $\varphi = 1$. Since $\varphi \mapsto -\cot(\pi\varphi)$ is strictly increasing on $(0, 1)$, comparing slopes and comparing phases is the same thing: $M(F) \leqslant M(E)$ if and only if $\varphi(F) \leqslant \varphi(E)$, and likewise for strict inequalities. We shall mostly work with phases; the rank R, the degree D and the slope M reappear in the proof of existence of HN filtrations, where the geometry of the plane is more conveniently expressed through them.

Let Z be a stability function on $\mathcal{A}$. A nonzero object $E \in \mathcal{A}$ is called $(\mathcal{A}, Z)$-*semistable* if

(AZ-SS) $\varphi(F) \leqslant \varphi(E)$ for every nontrivial subobject $F \subset E$,

and $(\mathcal{A}, Z)$-*stable* if

(AZ-S) $\varphi(F) < \varphi(E)$ for every nontrivial proper subobject $F \subset E$.

When $\mathcal{A}$ and Z are clear from the context we simply speak of semistable and stable objects.

1.3. **Harder–Narasimhan filtrations and stability conditions.** Let Z be a stability function on $\mathcal{A}$ and let $E \in \mathcal{A}$ be a nonzero object. A *Harder–Narasimhan filtration* (*HN filtration* for short) of E with respect to $(\mathcal{A}, Z)$ is a chain of subobjects

$$0 = E_0 \subset E_1 \subset \ldots \subset E_{n-1} \subset E_n = E$$

such that

- the factors $A_i := E_i/E_{i-1}$, $i = 1, \ldots, n$, are nonzero $(\mathcal{A}, Z)$-semistable objects;
- phases $\varphi(A_1) > \varphi(A_2) > \ldots > \varphi(A_n)$ are strictly decreasing.

The objects A_i are called the *HN factors* of E, and $\varphi^+(E) := \varphi(A_1)$, $\varphi^-(E) := \varphi(A_n)$ are called the *maximal* and the *minimal phase* of E.

$$\begin{array}{ccccccccccc} 0 = E_0 & \hookrightarrow & E_1 & \hookrightarrow & E_2 & \hookrightarrow & \cdots & \hookrightarrow & E_{n-1} & \hookrightarrow & E_n = E \\ & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow \\ & & A_1 & & A_2 & & & & A_{n-1} & & A_n \end{array}$$

Each column $E_{i-1} \hookrightarrow E_i \twoheadrightarrow A_i$ of the diagram is a short exact sequence; the horizontal arrows are the inclusions of the filtration.

An $(\mathcal{A}, Z)$-semistable object E has the HN filtration $0 \subset E$ of length $n = 1$; conversely, if an HN filtration of E has length 1, then E is $(\mathcal{A}, Z)$-semistable. Uniqueness of HN filtrations will be proved below; until then the words "maximal phase" refer to a chosen filtration.

A *stability condition* on the abelian category $\mathcal{A}$ is a pair $(\mathcal{A}, Z)$, where $Z\colon K_0(\mathcal{A}) \to \mathbb{C}$ is a group homomorphism, such that

(AZ-SC1) Z is a stability function on $\mathcal{A}$, i.e. condition (Z) holds;
(AZ-SC2) (*Harder–Narasimhan property*) every nonzero object $E \in \mathcal{A}$ admits an HN filtration with respect to $(\mathcal{A}, Z)$.

The homomorphism Z is called the *central charge* of the stability condition.

Remark. Condition (AZ-SC2) does not follow from (AZ-SC1): in Example 1.7 we shall give a stability function for which some object has no nonzero semistable subobject at all, hence no HN filtration. Sufficient conditions for the existence of HN filtrations (noetherianity of $\mathcal{A}$ together with discreteness of the image of $\operatorname{Im} Z$, or finite length of $\mathcal{A}$) will be discussed separately.

1.4. **Basic properties.** Throughout this subsection Z is a stability function on $\mathcal{A}$.

Lemma 1.1 (see-saw)**.** *Let $0 = E_0 \subset E_1 \subset \ldots \subset E_m = E$ be a chain of subobjects with nonzero factors $B_j := E_j/E_{j-1}$, $j = 1, \ldots, m$. Then*

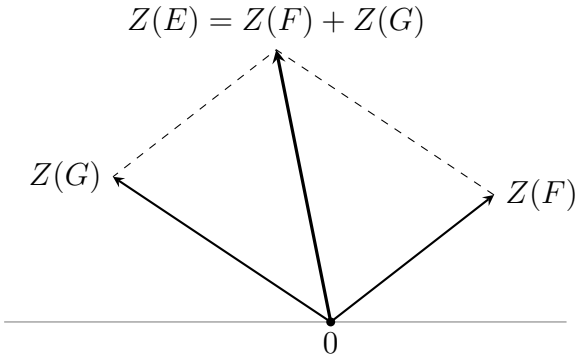
$$\min_j \varphi(B_j) \leqslant \varphi(E) \leqslant \max_j \varphi(B_j),$$

and equality holds on the left or on the right if and only if all the phases $\varphi(B_j)$ are equal.

In particular, for a short exact sequence $0 \to F \to E \to G \to 0$ with $F, G \neq 0$ the following are equivalent:

$$\varphi(F) < \varphi(E); \quad \varphi(E) < \varphi(G); \quad \varphi(F) < \varphi(G).$$

The same holds with $<$ replaced by $=$ or by $>$.



The picture shows the case of a short exact sequence $0 \to F \to E \to G \to 0$: since $Z(F)$ and $Z(G)$ both lie in $\mathbb{H}$, their sum $Z(E)$ lies in the sector between them, so the phase of E is caught between the phases of F and G.

Proof. By additivity, $Z(E) = \sum_j Z(B_j) = \sum_j r_j e^{i\pi\psi_j}$, where $r_j = m(B_j) > 0$ and $\psi_j = \varphi(B_j) \in (0, 1]$. Put $\psi := \max_j \psi_j$. Then

$$e^{-i\pi\psi} Z(E) = \sum_j r_j e^{i\pi(\psi_j - \psi)}, \qquad \psi_j - \psi \in (-1, 0],$$

so every summand has non-positive imaginary part, with equality if and only if $\psi_j = \psi$, in which case the summand is the positive real number r_j. Hence $\operatorname{Im}\left(e^{-i\pi\psi} Z(E)\right) \leqslant 0$, with equality if and only if all ψ_j equal ψ. On the other hand $e^{-i\pi\psi} Z(E) = m(E) e^{i\pi\theta}$ with $\theta := \varphi(E) - \psi \in (-1, 1)$, and $\sin(\pi\theta) \leqslant 0$ on this interval exactly when $\theta \in (-1, 0]$, with $\sin(\pi\theta) = 0$ exactly when $\theta = 0$. This proves the upper bound and its equality case; the lower bound is obtained in the same way with $\min_j \psi_j$ in place of ψ.

For a short exact sequence apply the above to the chain $0 \subset F \subset E$ with factors F, G: if, say, $\varphi(F) \leqslant \varphi(G)$, then $\varphi(F) \leqslant \varphi(E) \leqslant \varphi(G)$ and $\varphi(E)$ equals one of the endpoints only if $\varphi(F) = \varphi(G)$; the case $\varphi(F) \geqslant \varphi(G)$ is symmetric. □

Lemma 1.2 (quotient characterization)**.** *A nonzero object E is $(\mathcal{A}, Z)$-semistable if and only if $\varphi(E) \leqslant \varphi(G)$ for every nonzero quotient $E \twoheadrightarrow G$, and $(\mathcal{A}, Z)$-stable if and only if $\varphi(E) < \varphi(G)$ for every nonzero proper quotient.*

Proof. Nontrivial proper subobjects $F \subset E$ correspond to nonzero proper quotients $G = E/F$, and by Lemma 1.1 the inequality $\varphi(F) \leqslant \varphi(E)$ is equivalent to $\varphi(E) \leqslant \varphi(G)$, and similarly for strict inequalities. The subobject $F = E$ and the quotient $G = E$ impose no condition. □

Lemma 1.3 (Schur)**.** *Let E, F be $(\mathcal{A}, Z)$-semistable objects.*

(1) *If $\varphi(E) > \varphi(F)$, then* $\operatorname{Hom}(E, F) = 0$.
(2) *If $\varphi(E) = \varphi(F)$ and $f\colon E \to F$ is nonzero, then f is a monomorphism provided E is stable, an epimorphism provided F is stable, and an isomorphism provided both are stable.*

Proof. Let $f\colon E \to F$ be nonzero, with image $I \neq 0$ and kernel K. Since I is a quotient of E, Lemma 1.2 gives $\varphi(E) \leqslant \varphi(I)$; since $I \subset F$, semistability of F gives $\varphi(I) \leqslant \varphi(F)$. Thus $\varphi(E) \leqslant \varphi(F)$, which proves (1). In the situation of (2) all these inequalities are equalities. If E is stable and $K \neq 0$, then K is a nontrivial proper subobject of E (proper because $I \neq 0$) and, by Lemma 1.1 applied to

$$0 \to K \to E \to I \to 0,$$

$\varphi(E) = \varphi(I)$ forces $\varphi(K) = \varphi(E)$, contradicting stability; so $K = 0$. If F is stable, then $I \subset F$ with $\varphi(I) = \varphi(F)$ forces $I = F$. The last claim combines the two. □

Lemma 1.4. *If E and F are $(\mathcal{A}, Z)$-semistable of the same phase φ, then so is $E \oplus F$.*

Proof. First,

$$Z(E \oplus F) = Z(E) + Z(F)$$

is a sum of two vectors on the ray $\mathbb{R}_{>0} e^{i\pi\varphi}$, so $\varphi(E \oplus F) = \varphi$. Let $G \subset E \oplus F$ be nontrivial, and put $G' := \ker(G \to F) = G \cap E \subset E$ and $G'' := \operatorname{Im}(G \to F) \subset F$, so that $0 \to G' \to G \to G'' \to 0$ is exact. If both G' and G'' are nonzero, they have phases $\leqslant \varphi$ by semistability of E and F, and Lemma 1.1 gives $\varphi(G) \leqslant \varphi$; if one of them vanishes, G is isomorphic to the other and the same conclusion holds. □

Lemma 1.5. *Let E, F be nonzero objects admitting HN filtrations, with HN factors $A_1, \ldots, A_n$ and $B_1, \ldots, B_m$ respectively. If* $\operatorname{Hom}(E, F) \neq 0$*, then $\varphi^-(E) \leqslant \varphi^+(F)$; more precisely, $\varphi(A_i) \leqslant \varphi(B_j)$ for some i, j.*

Proof. Let $f\colon E \to F$ be nonzero, and let $0 = E_0 \subset \ldots \subset E_n = E$, $0 = F_0 \subset \ldots \subset F_m = F$ be the filtrations. Let i be minimal with $f|_{E_i} \neq 0$. Then $f|_{E_{i-1}} = 0$, so $f|_{E_i}$ factors through a nonzero map $g\colon A_i = E_i/E_{i-1} \to F$. Let j be minimal with $\operatorname{Im} g \subset F_j$. Then the composite $A_i \to F_j \to B_j = F_j/F_{j-1}$ is nonzero, since otherwise $\operatorname{Im} g \subset F_{j-1}$. By Lemma 1.3(1) applied to the semistable objects A_i and B_j, $\varphi(A_i) \leqslant \varphi(B_j)$. Finally $\varphi^-(E) \leqslant \varphi(A_i)$ and $\varphi(B_j) \leqslant \varphi^+(F)$. □

1.5. **Uniqueness of Harder–Narasimhan filtrations.**

Proposition 1.6. *Let Z be a stability function on $\mathcal{A}$ and let E be a nonzero object. If*

$$0 = E_0 \subset E_1 \subset \ldots \subset E_n = E \qquad \text{and} \qquad 0 = E'_0 \subset E'_1 \subset \ldots \subset E'_m = E$$

are two HN filtrations of E with respect to $(\mathcal{A}, Z)$, then $n = m$ and $E_i = E'_i$ as subobjects of E for all i. In particular the HN factors and the phases $\varphi^{\pm}(E)$ depend only on E.

Proof. Write $A_i := E_i/E_{i-1}$, $\varphi_i := \varphi(A_i)$ and $A'_j := E'_j/E'_{j-1}$, $\varphi'_j := \varphi(A'_j)$. We argue by induction on n.

Step 1 ($E_1 = E'_1$). Let j be minimal with $E_1 \subset E'_j$; it exists since $E'_m = E$. The composite $E_1 \to E'_j \to A'_j$ is nonzero, because otherwise $E_1 \subset E'_{j-1}$. Both $E_1 = A_1$ and A'_j are semistable, so Lemma 1.3(1) gives $\varphi_1 \leqslant \varphi'_j \leqslant \varphi'_1$. By symmetry $\varphi'_1 \leqslant \varphi_1$. Hence $\varphi_1 = \varphi'_1$, and then $\varphi'_j \geqslant \varphi'_1$ forces $j = 1$ because the phases φ'_j strictly decrease; thus $E_1 \subset E'_1$. By symmetry $E'_1 \subset E_1$, so $E_1 = E'_1$.

Step 2 (*induction*). If $n = 1$, then $E = E_1 = E'_1$, so $m = 1$ as well. If $n > 1$, consider $\bar{E} := E/E_1 = E/E'_1 \neq 0$. The images $\bar{E}_i := E_i/E_1$, $i \geqslant 1$, form a chain $0 = \bar{E}_1 \subset \ldots \subset \bar{E}_n = \bar{E}$ with factors $\bar{E}_i/\bar{E}_{i-1} \cong E_i/E_{i-1} = A_i$ for $i \geqslant 2$, hence an HN filtration of $\bar{E}$ of length $n-1$; similarly for $\bar{E}'_j := E'_j/E'_1$. By induction $n - 1 = m - 1$ and $\bar{E}_i = \bar{E}'_i$ for all $i \geqslant 1$. Since subobjects of $\bar{E}$ correspond bijectively to subobjects of E containing $E_1 = E'_1$, we get $E_i = E'_i$ for all i. □

Corollary 1.7. *Let $(\mathcal{A}, Z)$ be a stability condition and E a nonzero object. Then $\varphi^+(E) = \max\{\varphi(F) : 0 \neq F \subset E\}$, and the first step E_1 of the HN filtration is the unique largest subobject of E of phase $\varphi^+(E)$. Dually $\varphi^-(E) = \min\{\varphi(G) : E \twoheadrightarrow G \neq 0\}$.*

Proof. Let $0 \neq F \subset E$, with HN filtration $0 \subset F_1 \subset \ldots \subset F$. The composite $F_1 \hookrightarrow F \hookrightarrow E$ is a nonzero map from the semistable object F_1 to E, so Lemma 1.5 gives $\varphi^+(F) = \varphi(F_1) \leqslant \varphi^+(E)$, and $\varphi(F) \leqslant \varphi^+(F)$ by Lemma 1.1. The value $\varphi^+(E)$ is attained by $F = E_1$. Now suppose $\varphi(F) = \varphi^+(E)$. Then $\varphi(F) = \varphi^+(F)$, so by the equality case of Lemma 1.1 all HN factors of F have phase $\varphi^+(E)$, i.e. F is semistable of phase $\varphi^+(E)$. All HN factors of E/E_1 have phase $< \varphi^+(E)$, so the composite $F \to E \to E/E_1$ vanishes by Lemma 1.5, i.e. $F \subset E_1$. The dual statement follows by the same argument applied to quotients, using Lemma 1.2. □

1.6. **Existence of Harder–Narasimhan filtrations.** Recall that an abelian category $\mathcal{A}$ is *noetherian* if every ascending chain $F_1 \subset F_2 \subset \ldots$ of subobjects of any object stabilizes. Equivalently, every nonempty family of subobjects of an object has a maximal element with respect to inclusion. Throughout this subsection Z is a stability function on $\mathcal{A}$, and we write $R := \operatorname{Im} Z$ and $D := -\operatorname{Re} Z$, so that $R(F) \geqslant 0$ for all F, and $R(F) = 0$ implies $D(F) > 0$ for $F \neq 0$. We say that R *has discrete image* if $R(K_0(\mathcal{A}))$ is a discrete subgroup of $\mathbb{R}$; in that case either $R \equiv 0$, or $R(K_0(\mathcal{A})) = c\mathbb{Z}$ for a unique $c > 0$, and then $R(E) \in c\mathbb{Z}_{\geqslant 0}$ for every object E.

Two elementary identities will be used repeatedly. If $F, F' \subset E$ are subobjects, the sequence

$$0 \to F \cap F' \to F \oplus F' \to F + F' \to 0$$

is exact, hence

$$Z(F+F') + Z(F\cap F') = Z(F) + Z(F'), \tag{1}$$

and the same holds for R and D separately. If $F \subset E$ and $R(F) = R(E)$, then $R(E/F) = 0$, so $D(E/F) \geqslant 0$ and

$$D(F) \leqslant D(E) \qquad \text{whenever } F \subset E \text{ and } R(F) = R(E). \tag{2}$$

Lemma 1.8. *Assume that*

- $\mathcal{A}$ *is noetherian and*
- *that* R *has discrete image in* $\mathbb{R}$.

Then for every object E *there is a constant* $D_E \in \mathbb{R}$ *such that* $D(F) \leqslant D_E$ *for every subobject* $F \subset E$.

Proof. If $R \equiv 0$, then every $F \subset E$ satisfies $R(F) = R(E)$ and (2) gives $D(F) \leqslant D(E)$. Otherwise write $R(K_0(\mathcal{A})) = c\mathbb{Z}$ with $c > 0$ and argue by induction on $k(E) := R(E)/c \in \mathbb{Z}_{\geqslant 0}$.

If $k(E) = 0$, then $R(F) = 0 = R(E)$ for every $F \subset E$, and (2) gives $D(F) \leqslant D(E)$.

Let $k(E) > 0$ and assume the lemma for all objects E' with $k(E') < k(E)$. Suppose, for a contradiction, that there are subobjects $F_n \subset E$, $n \in \mathbb{Z}_{>0}$, with $D(F_n) \to \infty$. By (2) only finitely many of them can satisfy $R(F_n) = R(E)$, so after discarding these we may assume $R(F_n) < R(E)$ for all n. We construct indices $n_1 < n_2 < \dots$ such that the subobjects $G_k := F_{n_1} + \dots + F_{n_k}$ satisfy

$$R(G_k) < R(E) \qquad \text{and} \qquad D(G_k) > D(G_{k-1}) \quad (k \geqslant 2).$$

Put $n_1 := 1$. Given $n_1 < \dots < n_{k-1}$, the object G_{k-1} satisfies $k(G_{k-1}) < k(E)$, so by the induction hypothesis there is a constant D' with $D(F') \leqslant D'$ for all $F' \subset G_{k-1}$. For $n > n_{k-1}$, identity (1) and $G_{k-1} \cap F_n \subset G_{k-1}$ give

$$D(G_{k-1} + F_n) = D(G_{k-1}) + D(F_n) - D(G_{k-1} \cap F_n) \geqslant D(G_{k-1}) + D(F_n) - D',$$

which tends to ∞ as $n \to \infty$. Choose $n_k > n_{k-1}$ with

$$D(G_{k-1} + F_{n_k}) > \max\{D(E), D(G_{k-1})\}.$$

Then $R(G_{k-1}+F_{n_k}) < R(E)$ by (2), and $D(G_k) > D(G_{k-1})$ as required. Since $D(G_k) > D(G_{k-1})$, the inclusion $G_{k-1} \subset G_k$ is strict, so $G_1 \subsetneq G_2 \subsetneq \dots$ is an infinite strictly ascending chain of subobjects of E, contradicting noetherianity. □

Proposition 1.9. *Let* Z *be a stability function on* $\mathcal{A}$. *Assume that* $\mathcal{A}$ *is noetherian and that* $R = \operatorname{Im} Z$ *has discrete image. Then every nonzero object of* $\mathcal{A}$ *admits a Harder–Narasimhan filtration, i.e.* $(\mathcal{A}, Z)$ *is a stability condition.*

Proof. If $R \equiv 0$, then every nonzero object has phase 1, every nonzero object is semistable, and $0 \subset E$ is an HN filtration. So assume $R(K_0(\mathcal{A})) = c\mathbb{Z}$ with $c > 0$, and let $E \neq 0$.

Step 1 (*the part of rank zero*). The family of subobjects $F \subset E$ with $R(F) = 0$ is nonempty (it contains 0), so it has a maximal element T. If $F \subset E$ is any subobject with $R(F) = 0$, then $R(T + F) = R(T) + R(F) - R(T \cap F) = -R(T \cap F) \leqslant 0$ by (1), hence $R(T + F) = 0$ and $T + F = T$ by maximality, i.e. $F \subset T$. Thus T is the

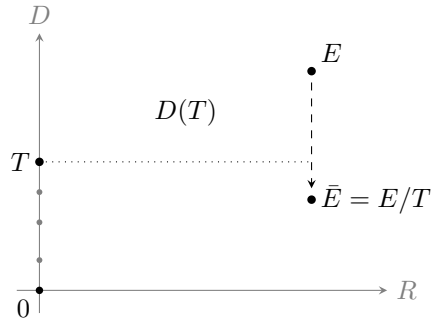
largest subobject of E of rank zero. If $T \neq 0$, it is semistable of phase 1, because all its nonzero subobjects have $R = 0$, hence phase 1. Moreover the quotient $\bar{E} := E/T$ has no nonzero subobject of rank zero: the preimage $F \subset E$ of such a subobject $\bar{F}$ satisfies $R(F) = R(T) + R(\bar{F}) = 0$, hence $F \subset T$ and $\bar{F} = 0$. Consequently every nonzero subobject of $\bar{E}$ has $R \geqslant c$, hence phase < 1. If $\bar{E} = 0$, then $E = T$ is semistable and we are done. Otherwise, if $0 = \bar{E}_0 \subset \bar{E}_1 \subset \ldots \subset \bar{E}_m = \bar{E}$ is an HN filtration of $\bar{E}$, then its factors have phases < 1, and the preimages

$$0 \subset T \subset \pi^{-1}(\bar{E}_1) \subset \ldots \subset \pi^{-1}(\bar{E}_m) = E$$

form an HN filtration of E (with the first step omitted if $T = 0$), since the factors are T (of phase 1) followed by the factors of the filtration of $\bar{E}$. Hence we may and do assume from now on that

(3) every nonzero subobject $F \subset E$ satisfies $R(F) \geqslant c$.

In the pictures accompanying this proof we use the coordinates $(R, D) = (\operatorname{Im} Z, -\operatorname{Re} Z)$, i.e. the plane of Z is drawn with the rank to the right and the degree upwards; a subobject $F \subset E$ is represented by the point $(R(F), D(F))$. Step 1 looks as follows: T is the highest point of the vertical axis occupied by a subobject of E, and passing from E to $\bar{E} = E/T$ shifts every point down by $D(T)$.



Step 2 (*the Harder–Narasimhan polygon*). For $r \in c\mathbb{Z}$ with $0 \leqslant r \leqslant R(E)$ let $\mathcal{S}_r$ be the set of subobjects $F \subset E$ with $R(F) = r$, and call r *occupied* if $\mathcal{S}_r \neq \emptyset$. The set Λ_E of occupied values is finite and contains 0 and $R(E)$. For $r \in \Lambda_E$ put

$$d(r) := \sup\{D(F) : F \in \mathcal{S}_r\},$$

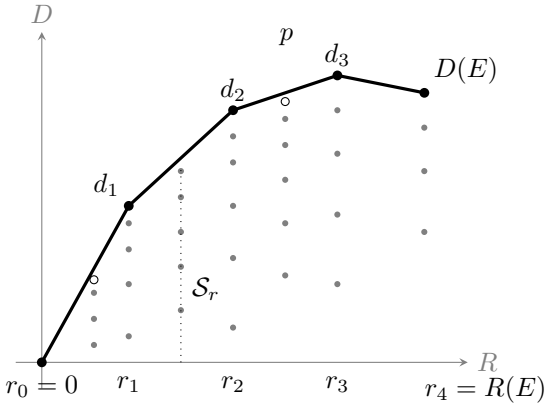
which is finite by Lemma 1.8. By (3) we have $\mathcal{S}_0 = \{0\}$ and $d(0) = 0$, and by (2) we have $d(R(E)) = D(E)$, attained by E itself. Let $p\colon [0, R(E)] \to \mathbb{R}$ be the upper concave envelope of the finite set of points $(r, d(r))$, $r \in \Lambda_E$, i.e. the smallest concave function with $p(r) \geqslant d(r)$ for all $r \in \Lambda_E$. It is piecewise linear, and its graph is the upper boundary of the convex hull of these points. Let $0 = r_0 < r_1 < \ldots < r_n = R(E)$ be the endpoints together with the points where the slope of p changes; we call them the *vertices*. They belong to Λ_E, satisfy $p(r_i) = d(r_i) =: d_i$, and the slopes

$$s_i := \frac{d_i - d_{i-1}}{r_i - r_{i-1}}, \qquad i = 1, \ldots, n,$$

are strictly decreasing, $s_1 > s_2 > \ldots > s_n$, because p is concave and changes slope at every r_i with $0 < i < n$. On $[r_{i-1}, r_i]$ the function p is affine with slope s_i. Finally, for every subobject $F \subset E$ we have $R(F) \in \Lambda_E$ and

$$D(F) \leqslant d(R(F)) \leqslant p(R(F)). \tag{4}$$

The picture below shows the sets $\mathcal{S}_r$ as vertical columns of points over the occupied values $r \in \Lambda_E$, the suprema $d(r)$ as the tops of the columns (a hollow circle marks a supremum which is not attained), the envelope p as the thick concave line, and the vertices $r_0, \ldots, r_n$ as the points where p bends.



Note that a non-vertex value may well be occupied (the column over the dotted line in the picture); what makes r_i a vertex is that p bends there.

Step 3 (*the suprema at the vertices are attained*). We claim that for every vertex r_i there is $F_i \in \mathcal{S}_{r_i}$ with $D(F_i) = d_i$. For $i = 0$ take $F_0 = 0$ and for $i = n$ take $F_n = E$. Let $0 < i < n$ and choose a number s with $s_i > s > s_{i+1}$. Consider the linear functional $\ell(x, y) := y - sx$ on $\mathbb{R}^2$ and put $M := \ell(r_i, d_i) = d_i - sr_i$. Since s lies between the slopes of p to the left and to the right of r_i, concavity gives $p(x) \leqslant d_i + s(x - r_i)$ for all $x \in [0, R(E)]$, with equality only for $x = r_i$. Together with (4) this shows

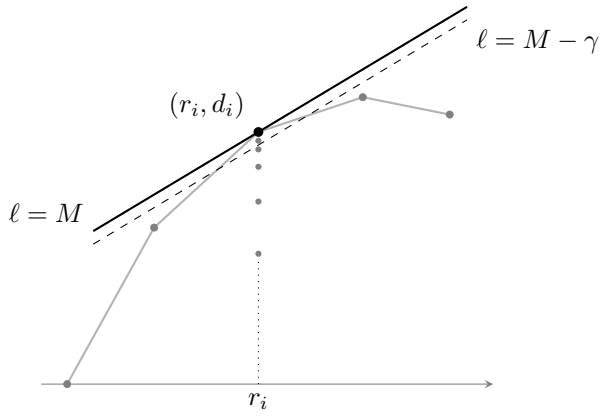
$$\ell\big(R(F), D(F)\big) \leqslant M \qquad \text{for every subobject } F \subset E, \tag{5}$$

and moreover, since Λ_E is finite,

$$\gamma := M - \max\big\{p(r) - sr : r \in \Lambda_E,\ r \neq r_i\big\} > 0,$$

so that $\ell(R(F), D(F)) > M - \gamma$ forces $R(F) = r_i$.

The picture shows the line of slope s through the vertex (r_i, d_i): it is a supporting line of the polygon, touching it only at that vertex, so all points $(R(F), D(F))$ lie on or below it, and the points of the other columns stay below the dashed parallel line, at vertical distance γ from the first one; only the points of the column $\mathcal{S}_{r_i}$ can come arbitrarily close to the solid line.



For brevity write $\ell(F) := \ell(R(F), D(F))$. Applying ℓ to (1) and using (5) for $F \cap F'$ we get, for all $F, F' \subset E$,

$$\ell(F + F') = \ell(F) + \ell(F') - \ell(F \cap F') \geqslant \ell(F) + \ell(F') - M. \tag{6}$$

Now choose $F^{(1)}, F^{(2)}, \ldots \in \mathcal{S}_{r_i}$ with $D(F^{(j)}) \to d_i$, and put $\varepsilon_j := M - \ell(F^{(j)}) = d_i - D(F^{(j)}) \geqslant 0$. Passing to a subsequence we may assume $\sum_j \varepsilon_j < \gamma$. Let $G_k := F^{(1)} + \ldots + F^{(k)}$. By induction on k, (6) gives

$$\ell(G_k) \geqslant \sum_{j=1}^{k} \ell(F^{(j)}) - (k-1)M = M - \sum_{j=1}^{k} \varepsilon_j > M - \gamma.$$

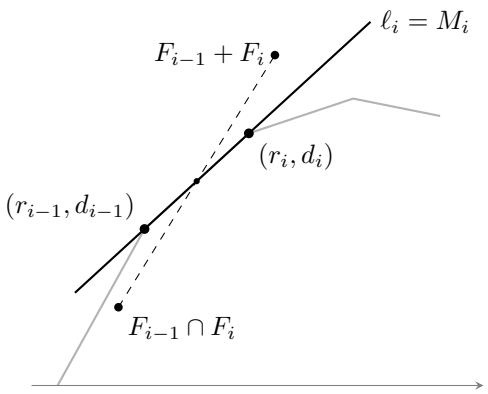
The chain $G_1 \subset G_2 \subset \ldots$ stabilizes by noetherianity, say $G_k = G$ for $k \geqslant N$. Then $\ell(G) > M - \gamma$, hence $R(G) = r_i$, and $G \supset F^{(j)}$ for every j. For $j \geqslant N$ the quotient $G/F^{(j)}$ has $R(G/F^{(j)}) = r_i - r_i = 0$, so $D(G/F^{(j)}) \geqslant 0$ and $D(G) \geqslant D(F^{(j)})$. Letting $j \to \infty$ we obtain $D(G) \geqslant d_i$, hence $D(G) = d_i$ by definition of d_i. Thus $F_i := G$ has the required property.

Step 4 (*the vertices form a chain*). Fix $1 \leqslant i \leqslant n$ and consider the functional $\ell_i(x, y) := y - s_i x$ and the number $M_i := \ell_i(r_{i-1}, d_{i-1}) = \ell_i(r_i, d_i)$. By concavity, $p(x) \leqslant d_{i-1} + s_i(x - r_{i-1})$ for all x, with equality exactly for $x \in [r_{i-1}, r_i]$, because the slope of p is $> s_i$ to the left of r_{i-1} and $< s_i$ to the right of r_i. Hence, by (4), $\ell_i(F) \leqslant M_i$ for every $F \subset E$, and $\ell_i(F) = M_i$ implies $r_{i-1} \leqslant R(F) \leqslant r_i$ and $D(F) = p(R(F))$. Apply (1) to F_{i-1} and F_i:

$$\ell_i(F_{i-1} \cap F_i) + \ell_i(F_{i-1} + F_i) = \ell_i(F_{i-1}) + \ell_i(F_i) = 2M_i.$$

Both summands on the left are $\leqslant M_i$, so both equal M_i.

The points of $F_{i-1} \cap F_i$ and $F_{i-1} + F_i$ are symmetric with respect to the midpoint of the edge from (r_{i-1}, d_{i-1}) to (r_i, d_i) by (1), and both must lie on or below the line $\ell_i = M_i$ carrying that edge; the dashed configuration is therefore impossible, and both points are forced onto the line, hence onto the edge.



Since $R(F_{i-1} \cap F_i) \leqslant R(F_{i-1}) = r_{i-1}$, the first equality forces $R(F_{i-1} \cap F_i) = r_{i-1}$ and $D(F_{i-1} \cap F_i) = p(r_{i-1}) = d_{i-1} = D(F_{i-1})$. Therefore the quotient $F_{i-1}/(F_{i-1} \cap F_i)$ has $R = 0$ and $D = 0$, i.e. $Z = 0$, so it is zero by (Z). Hence $F_{i-1} \subset F_i$. We have obtained a chain

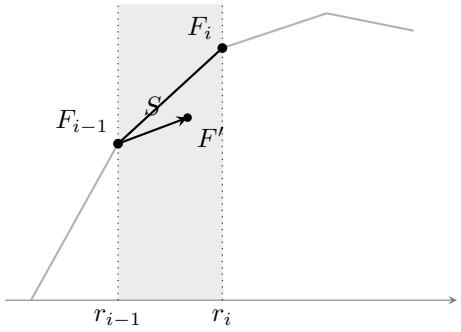
$$0 = F_0 \subset F_1 \subset \ldots \subset F_n = E.$$

Step 5 (*the factors are semistable with strictly decreasing phases*). Put $A_i := F_i/F_{i-1}$. Then $Z(A_i) = Z(F_i) - Z(F_{i-1})$, i.e. $R(A_i) = r_i - r_{i-1} > 0$ and $D(A_i) = d_i - d_{i-1}$, so $A_i \neq 0$ and its generalized slope is $M(A_i) = s_i$. Since $s_1 > \ldots > s_n$, the phases of the A_i strictly decrease. It remains to show that A_i is semistable. Let $S \subset A_i$ be a nontrivial subobject and $F' \subset F_i$ its preimage, so that $F_{i-1} \subsetneq F' \subset F_i$ and $Z(S) = Z(F') - Z(F_{i-1})$. Then $r_{i-1} \leqslant R(F') \leqslant r_i$, because $R(F'/F_{i-1}) \geqslant 0$ and $R(F_i/F') \geqslant 0$, and (4) together with the affine form of p on $[r_{i-1}, r_i]$ gives

$$D(F') \leqslant p(R(F')) = d_{i-1} + s_i\big(R(F') - r_{i-1}\big).$$

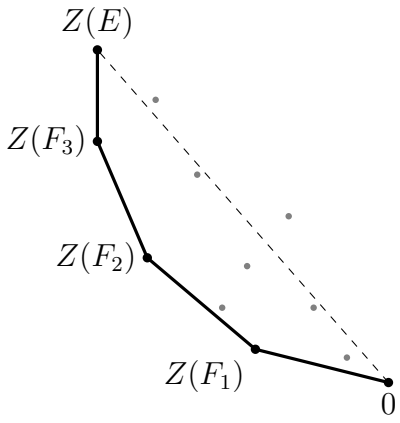
In other words $D(S) \leqslant s_i R(S)$.

The point of F' lies in the shaded strip $r_{i-1} \leqslant R \leqslant r_i$ and on or below the edge of the polygon; hence the vector from (r_{i-1}, d_{i-1}) to it, which represents $S = F'/F_{i-1}$, has slope at most s_i.



If $R(S) = 0$ this would give $D(S) \leqslant 0$, which is impossible for the nonzero object S; so $R(S) > 0$ and $M(S) = D(S)/R(S) \leqslant s_i = M(A_i)$, i.e. $\varphi(S) \leqslant \varphi(A_i)$. Thus A_i is semistable, and $0 = F_0 \subset \ldots \subset F_n = E$ is an HN filtration of E. $\square$

The proof shows that the HN filtration can be read off from the *Harder–Narasimhan polygon* of E: the part of the boundary of the convex hull of $\{Z(F) : F \subset E\}$ lying to the left of the segment from 0 to $Z(E)$. Its vertices are the points $Z(F_i)$ and its edges are the vectors $Z(A_i)$. In the picture below the grey dots are the values $Z(F)$ of other subobjects $F \subset E$, the dashed segment joins 0 to $Z(E)$, and the thick polygonal line is the HN polygon.



Corollary 1.10. *Let C be a smooth projective curve over a field $\Bbbk$ and $Z(E) = -\deg E + i \operatorname{rk} E$ on $\operatorname{Coh}(C)$. Then $(\operatorname{Coh}(C), Z)$ is a stability condition.*

Proof. $\operatorname{Coh}(C)$ is noetherian and $R = \operatorname{rk}$ takes values in $\mathbb{Z}$, so Proposition 1.9 applies. □

Remark. The proposition does not cover Example 1.7 in general, since there $\operatorname{Im} Z$ need not have discrete image; that example was treated by a direct finite-length argument. Bridgeland's original formulation replaces the two hypotheses by two chain conditions on phases, which cover both situations at once; noetherianity excludes infinite descending chains of quotients with decreasing phases, and discreteness of R excludes infinite descending chains of subobjects with increasing phases.

1.7. **Examples. Example.** Let C be a smooth projective curve over a field $\Bbbk$, $\mathcal{A} = \operatorname{Coh}(C)$, and

$$Z(E) := -\deg E + i \operatorname{rk} E.$$

This is a stability function: for a nonzero sheaf E either $\operatorname{rk} E > 0$, or E is a torsion sheaf, in which case $\deg E$ is the length of E, which is positive, so $Z(E) \in \mathbb{R}_{<0}$. For $\operatorname{rk} E > 0$ the generalized slope $M(E) = \deg E / \operatorname{rk} E$ is the usual slope, and torsion sheaves have $M = +\infty$; thus $(\mathcal{A}, Z)$-(semi)stability is slope (semi)stability, and semistable sheaves of positive rank are torsion-free. The Harder–Narasimhan property holds (Harder–Narasimhan), so $(\operatorname{Coh}(C), Z)$ is a stability condition. This is Corollary 1.10. ★

Example. Let X be a smooth projective variety of dimension $n \geqslant 2$ and ω an ample divisor class. The homomorphism $Z(E) := -\omega^{n-1} \cdot c_1(E) + i\, \omega^n \cdot \operatorname{rk} E$ on $K_0(\operatorname{Coh}(X))$ is not a stability function: for a nonzero sheaf T supported in codimension $\geqslant 2$ both $\operatorname{rk} T$ and $c_1(T)$ vanish, so $Z(T) = 0 \notin \mathbb{H}$. One can show that no homomorphism factoring through the Chern character is a stability function on $\operatorname{Coh}(X)$; this is the reason for passing to other hearts in the derived category. ★

Example. Let A be a finite-dimensional algebra over a field $\Bbbk$, $\mathcal{A} = \operatorname{mod}\text{-}A$ the category of finite-dimensional right A-modules, and $S_1, \dots, S_m$ representatives of the isomorphism classes of simple modules. Every $M \in \mathcal{A}$ has a composition series, and by the Jordan–Hölder theorem the multiplicity $[M : S_j]$ of S_j among its factors is well defined and additive in short exact sequences. Hence

$$K_0(\mathcal{A}) \longrightarrow \mathbb{Z}^m, \qquad [M] \longmapsto \big([M : S_1], \dots, [M : S_m]\big)$$

is a well-defined homomorphism; it is surjective, and it is injective because every class $[M]$ equals $\sum_j [M : S_j][S_j]$. So $[S_1], \dots, [S_m]$ form a basis of $K_0(\mathcal{A}) \cong \mathbb{Z}^m$, and for any $z_1, \dots, z_m \in \mathbb{C}$ there is a unique homomorphism Z with $Z(S_j) = z_j$, namely $Z(M) = \sum_j [M : S_j] z_j$.

Now let $z_1, \dots, z_m \in \mathbb{H}$. Then Z is a stability function, since $Z(M)$ is a sum of vectors in $\mathbb{H}$ with non-negative coefficients, not all zero. We claim that $(\mathcal{A}, Z)$ is a stability condition. Let $M \neq 0$; we construct an HN filtration by induction on the length $\ell(M)$.

For a subobject $N \subset M$ one has $0 \leqslant [N : S_j] \leqslant [M : S_j]$, so the classes $[N]$, hence the phases $\varphi(N)$, range over a finite set. Let φ_1 be the maximal phase of a nontrivial subobject, and among the nontrivial subobjects of phase φ_1 choose M_1 of maximal length. Then M_1 is semistable: its subobjects are subobjects of M, hence have phase $\leqslant \varphi_1$. Moreover every nontrivial $N \subset M$ with $\varphi(N) = \varphi_1$ satisfies $N \subset M_1$. Indeed, from the exact sequence

$$0 \to M_1 \cap N \to M_1 \oplus N \to M_1 + N \to 0$$

and Lemma 1.4, $\varphi(M_1 \oplus N) = \varphi_1$, while $\varphi(M_1 \cap N) \leqslant \varphi_1$ if $M_1 \cap N \neq 0$; so Lemma 1.1 gives $\varphi(M_1 + N) \geqslant \varphi_1$, hence $= \varphi_1$ by maximality of φ_1. By maximality of $\ell(M_1)$, $\ell(M_1 + N) \leqslant \ell(M_1)$, so $M_1 + N = M_1$. (If $M_1 \cap N = 0$, then $M_1 + N \cong M_1 \oplus N$ has phase φ_1 and larger length, which is impossible, so this case does not occur.)

Finally, every nontrivial subobject $N' \subset M/M_1$ has $\varphi(N') < \varphi_1$: if $N \subset M$ is its preimage, then $0 \to M_1 \to N \to N' \to 0$ with $\varphi(N) \leqslant \varphi_1 = \varphi(M_1)$, so Lemma 1.1 gives $\varphi(N') \leqslant \varphi(N) \leqslant \varphi_1$, and $\varphi(N') = \varphi_1$ would force $\varphi(N) = \varphi_1$, hence $N \subset M_1$ and $N' = 0$. Since $\ell(M/M_1) < \ell(M)$, by induction M/M_1 has an HN filtration; its first factor has phase $< \varphi_1$ by what we just proved, so the preimage of this filtration in M, preceded by M_1, is an HN filtration of M. If $M_1 = M$ there is nothing to do. ★

Remark. The argument of Example 1.7 uses only that $\mathcal{A}$ has finite length and that the classes of subobjects of a given object form a finite set; it applies verbatim to any abelian category of finite length with finitely many simple objects.

Example.[a stability function without the Harder–Narasimhan property] Let $\mathcal{A}$ be the category of finitely generated $\mathbb{Z}$-graded modules over the polynomial ring $\Bbbk[x]$ with $\deg x = 1$, and graded homomorphisms of degree 0. For $M = \bigoplus_n M_n \in \mathcal{A}$ the pieces M_n vanish for $n \ll 0$ and have bounded dimension, so

$$Z(M) := \sum_{n \in \mathbb{Z}} \big(-2^{-n} + i\, 4^{-n} \big) \dim_{\Bbbk} M_n$$

converges absolutely. Short exact sequences in $\mathcal{A}$ are exact in each degree, so Z is additive, and $\operatorname{Im} Z(M) > 0$ for $M \neq 0$; thus Z is a stability function.

Consider $E := \Bbbk[x]$, generated in degree 0. Its graded submodules are exactly the ideals $(x^j) \cong \Bbbk[x](-j)$, $j \geqslant 0$, and 0. For $j \geqslant 0$,

$$Z\big((x^j)\big) = \sum_{n \geqslant j} \big(-2^{-n} + i\, 4^{-n} \big) = -2^{1-j} + i\, \tfrac{4}{3}\, 4^{-j},$$

so $\tan\big(\pi - \pi\varphi((x^j))\big) = \frac{2}{3} \cdot 2^{-j}$ is strictly decreasing in j; hence the phases $\varphi((x^j))$ strictly increase with j. Consequently no nonzero subobject of E is semistable: (x^j) has the subobject (x^{j+1}) of strictly larger phase. But the first step E_1 of an HN filtration of E would be a nonzero semistable subobject of E. Hence E has no HN filtration, and $(\mathcal{A}, Z)$ is not a stability condition. ★